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23. Minimization of DFA

5.2

Minimization of DFA is required to obtain the minimal version of any DFA which consists of the minimum number of states possible.

DFA 5 states

4 states

0 0 0 0 0
└─┘

Equivalent

Two states 'A' and 'B' are said to be equivalent if

$$\left\{ \begin{array}{l} \delta(A, x) \rightarrow F \\ \delta(B, x) \rightarrow F \end{array} \right. \text{ and } \text{OR} \left\{ \begin{array}{l} \delta(A, x) \not\rightarrow F \\ \delta(B, x) \not\rightarrow F \end{array} \right.$$

where 'x' is any input string



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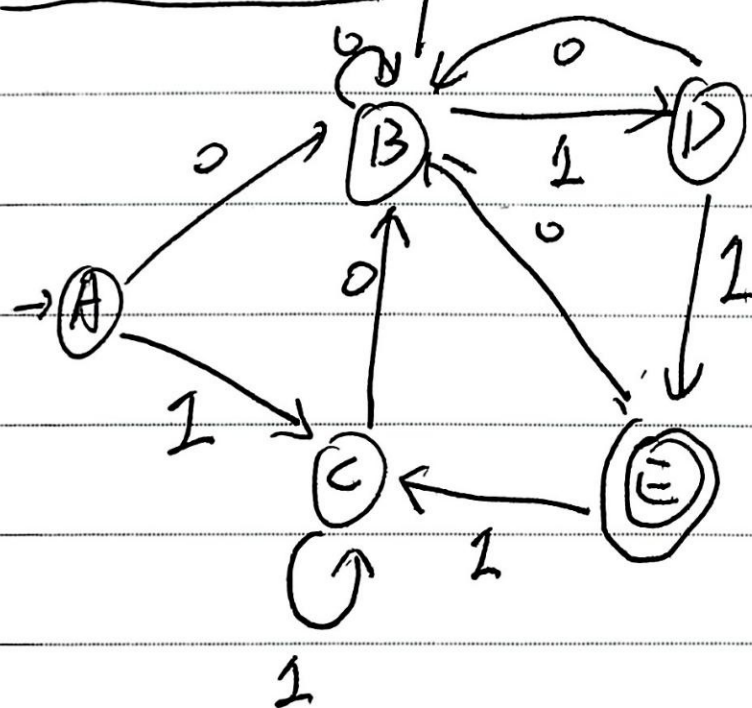
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If $|x| = i$, then A and B are said to be i equivalent

length

21, Example 1



$\Rightarrow$

	0	1
<u>A</u>	<u>B</u>	<u>C</u>
<u>B</u>	<u>B</u>	<u>D</u>
<u>C</u>	B	C
D	B	E
<u>E</u>	B	C

0 Equivalence: $\{A, B, C, D\}, \{E\}$

1 Equivalence: $\{A, B, C\}, \{D\}, \{E\}$ A, B ✓

2 Equivalence: ~~$\{A, B\}$~~ A, C ✓

$\{A, C\}, \{B\}, \{D\}, \{E\}$ C, D X

3 Equivalence: $\{A, C\}, \{B\}, \{D\}, \{E\}$ $\rightarrow$ the same $\rightarrow$ stop

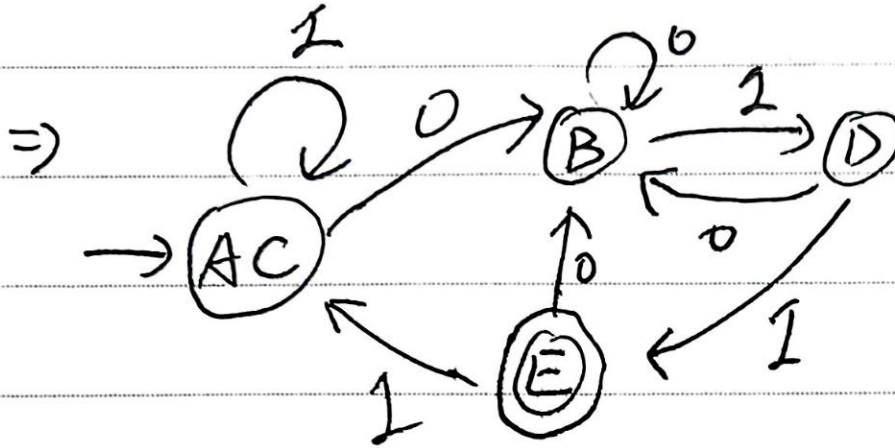


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so A, C can be combined to one state



	0	1
→ A	B	C
B	B	D
C	B	C
D	B	E
(E)	B	C



22. Example 2 of Minimum DFA

	0	1	
→ q_0	q_1	q_5	0 Equivalence:
q_1	q_6	q_2	$\{q_0, q_1, q_3, q_4, q_5, q_6, q_7\}, \{q_2\}$
(q_2)	q_0	q_2	1 Equivalence:
q_3	q_2	q_6	$\{q_0, q_4, q_6\}$
q_4	q_7	q_5	$\{q_1, q_7\}$
q_5	q_2	q_6	$\{q_3, q_5\}$
q_6	q_6	q_4	$\{q_2\}$
q_7	q_6	q_2	2. Equivalence:
			$\{q_0, q_4\}$ $\{q_6\}$
			$\{q_1, q_7\}$
			$\{q_3, q_5\}$
			$\{q_2\}$

3. Equivalence:

same as 2 equivalence → stop

so combine $\{q_0, q_4\}, \{q_1, q_7\}, \{q_3, q_5\}$



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0	1	0	1
→ {20, 24}	{21, 27}	{23, 25}	
{26}	{26}	{20, 24}	
{21, 27}	{26}	{22}	
{23, 25}	{22}	{26}	
{22}	{20, 24}	{22}	

4

am

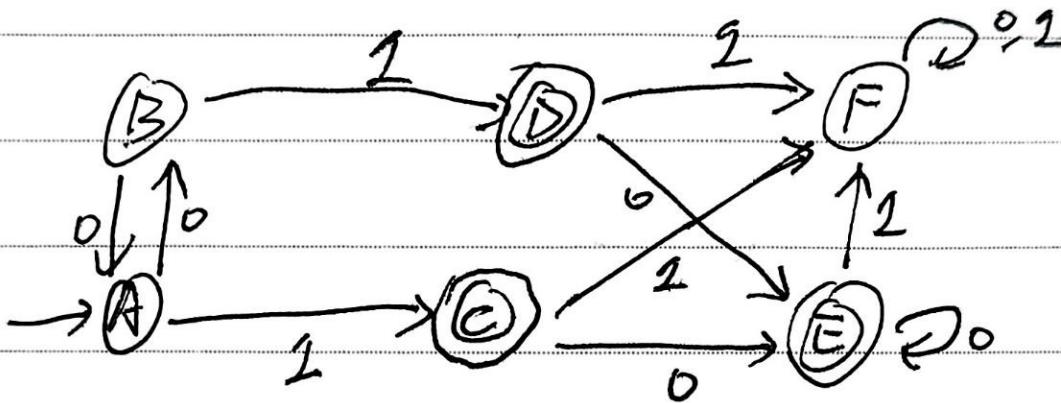
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123. Example of Minimization of DFA - Ex 3

for more than one Final state

Minimize the following DFA:



	0	1
→ A	B	C
B	A	D
C	E	F
D	E	F
E	E	F
F	F	F

0-Equivalence:

$\{A, B, F\}, \{C, D, E\}$

1-Equivalences

$\{A, B\}, \{F\}$

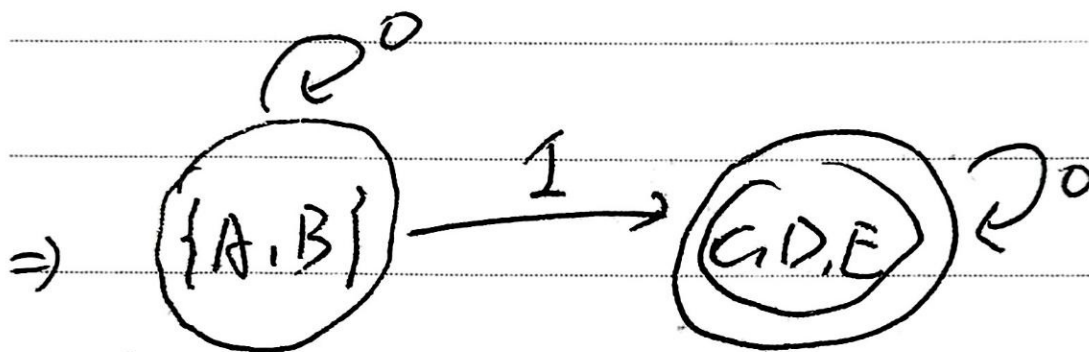
$\{C, D, E\}$

2 - Equivalence:

$\{A, B\}, \{F\}$

$\{C, D, E\} \Rightarrow \text{stop}$

	0	1
$\rightarrow \{A, B\}$	$\{A, B\}$	$\{C, D, E\}$
$\{F\}$	$\{F\}$	$\{F\}$
$\nrightarrow \{C, D, E\}$	$\{C, D, E\}$	$\{F\}$



diagram



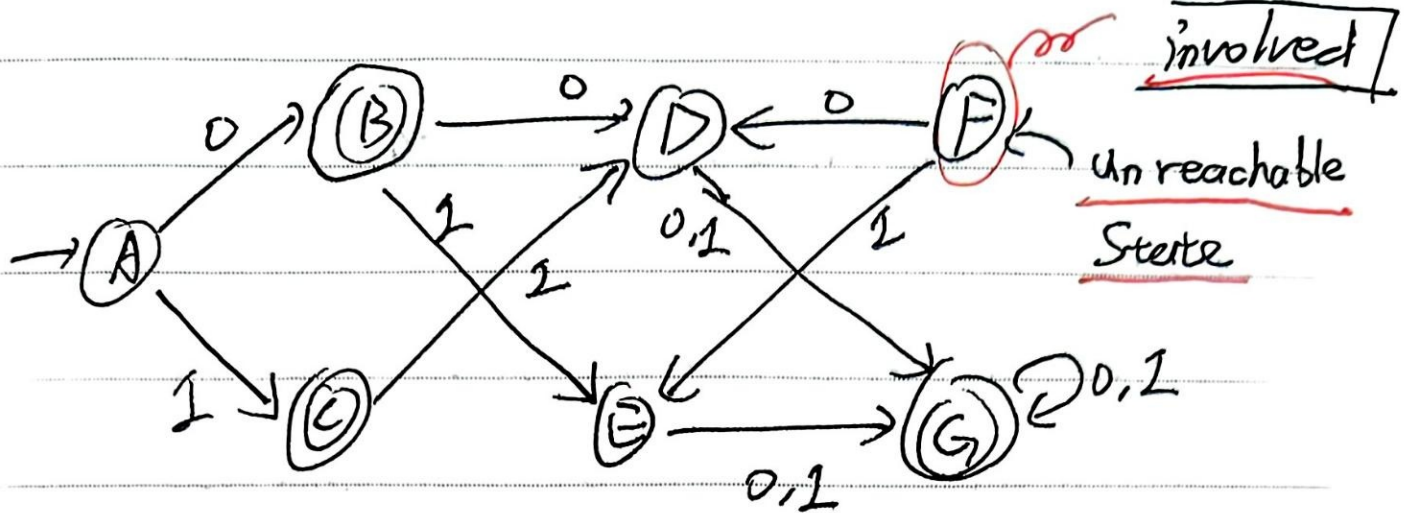


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24. Example 4 - there are unreachable States



just remove unreachable state

0 1 0-Equivalence: {A, D, E}, {B, C, G}

→ A B C 1-Equivalence: {A, D, E}, {B, C}, {G}

(B) D E 2-Equivalence: {A}, {D, E}, {B, C}, {G}

(C) E D 3-Equivalence: {A}, {D, E}, {B, C}, {G}

D G G

E G G mini-DEA 0 1

(G) G G → {A} {B, C} {B, C}

{D, E} {G} {G}

{B, C} {D, E} {D, E}

{G} {G} {G}

